

Finding the Best Rational Approximation of a Floating-Point Number

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Nov 18, 2025

Abstract

Let x be the absolute value of a given finite floating-point number, and t the given tolerance (absolute error tolerated). We seek the best rational approximation $\frac{p}{q}$ for x such that $\left| \frac{p}{q} - x \right| \leq t$.

Convergents

A basic approach would be to compute the convergents $\frac{p_n}{q_n}$ of the regular continued fraction representation of x using the Euclidean algorithm ie.

- Initial conditions :

$$\begin{aligned}(p_{-2}, p_{-1}) &= (0, 1) \\ (q_{-2}, q_{-1}) &= (1, 0)\end{aligned}$$

$$\begin{aligned}x_0 &= x \\ a_0 &= \lfloor x_0 \rfloor \\ r_0 &= x_0 - a_0\end{aligned}$$

- Recursive step for $n > 0$:

$$\begin{aligned}x_n &= \frac{1}{r_{n-1}} \\ a_n &= \lfloor x_n \rfloor \\ r_n &= x_n - a_n\end{aligned}$$

- Recurrence relation for $n \geq 0$:

$$\begin{aligned}p_n &= a_n \cdot p_{n-1} + p_{n-2} \\ q_n &= a_n \cdot q_{n-1} + q_{n-2}\end{aligned}$$

- Predicate :

$$\left| \frac{p_n}{q_n} - x \right| > t$$

until we obtain a convergent $\frac{p_n}{q_n}$ that satisfies $\left| \frac{p_n}{q_n} - x \right| \leq t$.

However at this point, while standard continued fractions generate “best approximations of the second kind”, $\frac{p_n}{q_n}$ is not necessarily the best rational approximation for x **given t** , that is, **the one satisfying the inequality that has the smallest denominator** (ie. any approximation with a larger denominator would be an unnecessary use of precision).

Optimal Semiconvergent

Actually the best rational approximation we are after is one of the first kind, a semiconvergent $\frac{p}{q}$ such that $\left| \frac{p}{q} - x \right| \leq t$, with :

$$\begin{aligned} p_{n-1} &< p \leq p_n \\ q_{n-1} &< q \leq q_n \end{aligned}$$

where q and p are minimized, which amounts to find the smallest integer value a such that $0 < a \leq a_n$ for which $\left| \frac{p}{q} - x \right| \leq t$, from the recurrence relation :

$$\begin{aligned} p &= a \cdot p_{n-1} + p_{n-2} \\ q &= a \cdot q_{n-1} + q_{n-2} \end{aligned}$$

The magnitude of the error of $\frac{p}{q}$ is :

$$\left| \frac{p}{q} - x \right| = \left| \frac{a \cdot p_{n-1} + p_{n-2}}{a \cdot q_{n-1} + q_{n-2}} - x \right|$$

$$\left| \frac{p - x \cdot q}{q} \right| = \left| \frac{a \cdot p_{n-1} + p_{n-2} - x(a \cdot q_{n-1} + q_{n-2})}{a \cdot q_{n-1} + q_{n-2}} \right|$$

$$\left| \frac{p - x \cdot q}{q} \right| = \left| \frac{a \cdot (p_{n-1} - x \cdot q_{n-1}) + (p_{n-2} - x \cdot q_{n-2})}{a \cdot q_{n-1} + q_{n-2}} \right|$$

$$\text{let } e_n = p_n - x \cdot q_n$$

$$\left| \frac{p - x \cdot q}{q} \right| = \left| \frac{a \cdot e_{n-1} + e_{n-2}}{a \cdot q_{n-1} + q_{n-2}} \right|$$

NB. e_{n-1} and e_{n-2} are the signed errors of the previous two convergents multiplied by their respective denominators. As a increases, the denominator $a \cdot q_{n-1} + q_{n-2}$ increases and the absolute error decreases monotonically.

We want the smallest positive integer value a for which the absolute error is less than or equal to t :

$$\left| \frac{a \cdot e_{n-1} + e_{n-2}}{a \cdot q_{n-1} + q_{n-2}} \right| \leq t$$

Let m a positive number such that $\left| \frac{m \cdot e_{n-1} + e_{n-2}}{m \cdot q_{n-1} + q_{n-2}} \right| = t$. This equation has 2 solutions for m , one on each side of x on the number line, where the error equals $\pm t$:

$$\frac{m \cdot e_{n-1} + e_{n-2}}{m \cdot q_{n-1} + q_{n-2}} = \pm t$$

$$m \cdot e_{n-1} + e_{n-2} = \pm t \cdot (m \cdot q_{n-1} + q_{n-2})$$

$$m \cdot e_{n-1} = -e_{n-2} \pm t \cdot m \cdot q_{n-1} \pm t \cdot q_{n-2}$$

$$m \cdot e_{n-1} \mp t \cdot m \cdot q_{n-1} = -e_{n-2} \pm t \cdot q_{n-2}$$

$$m = \frac{-e_{n-2} \pm t \cdot q_{n-2}}{e_{n-1} \mp t \cdot q_{n-1}}$$

Since e_{n-1} and e_{n-2} have opposite signs, we have :

$$m_1 = \frac{|e_{n-2}| - t \cdot q_{n-2}}{|e_{n-1}| + t \cdot q_{n-1}}$$

$$m_2 = \frac{|e_{n-2}| + t \cdot q_{n-2}}{|e_{n-1}| - t \cdot q_{n-1}}$$

Since $m_1 < m_2$, the smallest integer a for which the absolute error of $\frac{p}{q}$ is less than or equal to t equals $\lceil m_1 \rceil$:

$$a = \left\lceil \frac{|e_{n-2}| - t \cdot q_{n-2}}{|e_{n-1}| + t \cdot q_{n-1}} \right\rceil$$

Floating-point arithmetic and precision issues

Floating-point operations cannot accurately represent true arithmetic operations, so implementing the recursive step mentioned above as it is using floating-point arithmetic would be inexact : as the result of $1/r_{n-1}$ is subject to roundoff error, $x_n = \frac{1}{r_{n-1}}$ translates to $x_n \approx \frac{1}{r_{n-1}}$, and $\lfloor x_n \rfloor$ could be off by ± 1 (and thus the value of a_n and r_n). For the same reason, the result of the ceil division used to find the smallest a could be off as well, and the comparison between $|p_n/q_n - x|$ and t might be wrong. In other words, we need to avoid floating-point division roundoff error, or at least be able to measure it.

There are two options to address this :

- use twice the precision of x to represent the involved variables and for every operations (using Veltkamp/Dekker algorithms if x cannot be represented with a higher precision number type in the language used for the implementation).
- or prevent intermediate rounding when computing a_n and keep track of the corresponding error term e_n at each step n .

Preventing intermediate rounding

The error term also has a recurrence relation :

$$\begin{aligned}
 e_n &= p_n - x \cdot q_n \\
 &= a_n \cdot p_{n-1} + p_{n-2} - x \cdot (a_n \cdot q_{n-1} + q_{n-2}) \\
 &= a_n \cdot (p_{n-1} - x \cdot q_{n-1}) + p_{n-2} - x \cdot q_{n-2} \\
 e_n &= a_n \cdot e_{n-1} + e_{n-2}
 \end{aligned}$$

We can write this as $e_{n-2} = -a_n \cdot e_{n-1} + e_n$, which means, since $|e_n| < |e_{n-1}|$, that **$-a_n$ and e_n are respectively the quotient and remainder of $\frac{e_{n-2}}{e_{n-1}}$** .

So we can compute e_n as the remainder of $\frac{e_{n-2}}{e_{n-1}}$ to obtain an error term which is coherent with the value of a_n , precisely because we now have $a_n = \left\lfloor \frac{e_{n-2} - e_n}{e_{n-1}} \right\rfloor$ which remains error-free using floating-point arithmetic. The same approach should be used to avoid intermediate rounding when calculating the smallest a .

Another advantage of doing this is that we can check if the error term matches the tolerance before even having to produce the corresponding convergents for n . In order to do that, we also need to maintain a tolerance term t_n that is comparable to e_n . Since $\left| \frac{p_n}{q_n} - x \right| \leq t$ is equivalent to $|e_n| \leq t \cdot q_n$, we define $t_n = t \cdot q_n$ in the initial conditions and use the recurrence relation $t_n = a_n \cdot t_{n-1} + t_{n-2}$ in the next steps to avoid relying on q_n for the comparison.

NB. The exactness of t_n after some iterations depends on the precision required by t . Actually, inaccuracies can occur when the binary representation of t occupies most of the bits of its mantissa. In practice, t most likely is (should be) the unit of least precision of x , in which case t_n remains exact.

Best rational approximation

By maintaining an exact error term e_n and a scaled tolerance t_n , we can eliminate floating-point roundoff errors and avoid maintaining the tail (x_n and r_n) of the continued fraction. The optimized algorithm is based on the following :

- Initial conditions :

$$\begin{aligned}
 (p_{-2}, p_{-1}) &= (0, 1) \\
 (q_{-2}, q_{-1}) &= (1, 0) \\
 (e_{-2}, e_{-1}) &= (-x, 1) \\
 (t_{-2}, t_{-1}) &= (t, 0)
 \end{aligned}$$

- Recursive step for $n \geq 0$:

$$e_n = \text{rem}(e_{n-2}, e_{n-1})$$

$$a_n = \left\lfloor \frac{|e_{n-2} - e_n|}{e_{n-1}} \right\rfloor$$

- Recurrence relations for $n \geq 0$:

$$p_n = a_n \cdot p_{n-1} + p_{n-2}$$

$$q_n = a_n \cdot q_{n-1} + q_{n-2}$$

$$t_n = a_n \cdot t_{n-1} + t_{n-2}$$

- Predicate :

$$|e_n| > t_n$$

where $\text{rem}(x, y)$ is a remainder function that takes the sign of x , for example :

- C : `fmod(x, y)`
- JavaScript : `x % y`
- Julia: `x % y, rem(x, y, RoundToZero)`
- Python: `math.fmod(x, y)` (% operator is fine if working with absolute values for e_n)

Once $|e_n| \leq t_n$, we know $\frac{p_n}{q_n}$ satisfies the tolerance.

If $a_n = 1$, then $\frac{p_n}{q_n}$ is already optimal. Otherwise, we find the optimal semi-convergent $\frac{p}{q}$ by calculating the smallest valid coefficient a using the error and tolerance terms, then compute the final numerator and denominator :

$$a = \left\lceil \frac{|e_{n-2}| - t_{n-2}}{|e_{n-1}| + t_{n-1}} \right\rceil$$

$$p = a \cdot p_{n-1} + p_{n-2}$$

$$q = a \cdot q_{n-1} + q_{n-2}$$